Requirements Traceability Matrix

Writer's Creative Ideas Information System

Academic / Engineering Project

Revised client-server baseline

This matrix links each requirement to its acceptance criterion, applicable Given–When–Then (BDD) scenario, planned Jira ticket identifiers, and available verification evidence. Scenario references use the numbering and titles in Section 5 of `spec/srs.tex`. Architectural requirements cover the mobile client, REST API resources, and offline synchronization. Addendum 02 introduces REQ-F-038 and REQ-F-039 but defines no corresponding acceptance criteria or dedicated BDD scenarios; this remains an explicit traceability gap. API-GATE results below are aggregate mock/API evidence, not requirement-by-requirement acceptance results. References to `mock-api/server.js` identify a test mock, not production implementation code.

Ticket responsibility and authority. The AI agent handles execution and generation for `WNT-05`, `WNT-06`–`WNT-12`, `WNT-15` (acceptance test execution with at least 85% coverage), and `WNT-16`. For `WNT-01`, `WNT-02`, and architectural details in `WNT-09`–`WNT-12`, work is collaborative: the agent proposes options and maps requirements; the human reviews and adjusts. Human-only authority applies to `WNT-03`, `WNT-04` (Gate 1), `WNT-13`, `WNT-14` (Gate 2), and `WNT-17` (Gate 3). Design validation, implementation verification, and all GO/NO-GO decisions remain exclusively with the human.

Requirement ID	Acceptance Criterion	BDD Scenario	Ticket	Test	Code	Documentation
REQ-F-001	AC-F-001	S7: Offline Use and Persistence	`WNT-06`			
REQ-F-002	AC-F-002	S2: Create an Entry with All Details	`WNT-06`			
REQ-F-003	AC-F-003	S1: Generate a Display Title; S2: Create an Entry with All Details	`WNT-06`			
REQ-F-004	AC-F-004	S2: Create an Entry with All Details; S8: Create and Structure a General Note	`WNT-06`			
REQ-F-005	AC-F-005	S8: Create and Structure a General Note	`WNT-06`			
REQ-F-006	AC-F-006	S2: Create an Entry with All Details; S8: Create and Structure a General Note	`WNT-06`			

Requirement ID	Acceptance Criterion	BDD Scenario	Ticket	Test	Code	Documentation
REQ-F-007	AC-F-007	S2: Create an Entry with All Details; S8: Create and Structure a General Note	WNT-06			
REQ-F-008	AC-F-008	S10: Delete an Entry	WNT-08			
REQ-F-009	AC-F-009	S2: Create an Entry with All Details; S8: Create and Structure a General Note	WNT-07			
REQ-F-010	AC-F-010	S3: Chapter and Book Associations	WNT-06			
REQ-F-011	AC-F-011	S3: Chapter and Book Associations	WNT-07			
REQ-F-012	AC-F-012	S3: Chapter and Book Associations; S6: Export Book Project in Chapter Order	WNT-07			
REQ-F-013	AC-F-013	S3: Chapter and Book Associations; S8: Create and Structure a General Note	WNT-07			
REQ-F-014	AC-F-014	S3: Chapter and Book Associations	WNT-07			
REQ-F-015	AC-F-015	S3: Chapter and Book Associations; S8: Create and Structure a General Note	WNT-07			
REQ-F-016	AC-F-016	S3: Chapter and Book Associations	WNT-07			
REQ-F-017	AC-F-017	S3: Chapter and Book Associations	WNT-08			
REQ-F-018	AC-F-018	S3: Chapter and Book Associations	WNT-08			
REQ-F-019	AC-F-019	S4: Symmetric Entry Links	WNT-08			
REQ-F-020	AC-F-020	S4: Symmetric Entry Links	WNT-08			
REQ-F-021	AC-F-021	S4: Symmetric Entry Links	WNT-08			
REQ-F-022	AC-F-022	S5: Search and Combined Filters; S7: Offline Use and Persistence	WNT-09			
REQ-F-023	AC-F-023	S5: Search and Combined Filters	WNT-09			
REQ-F-024	AC-F-024	S5: Search and Combined Filters	WNT-09			
REQ-F-025	AC-F-025	S5: Search and Combined Filters	WNT-09			

Requirement ID	Acceptance Criterion	BDD Scenario	Ticket	Test	Code	Documentation
REQ-F-026	AC-F-026	S7: Offline Use and Persistence; S9: Export Success and Failure	WNT-02, WNT-10			
REQ-F-027	AC-F-027	S6: Export Book Project in Chapter Order	WNT-02, WNT-10			
REQ-F-028	AC-F-028	S6: Export Book Project in Chapter Order	WNT-02, WNT-10			
REQ-F-029	AC-F-029	S6: Export Book Project in Chapter Order	WNT-02, WNT-10			
REQ-F-030	AC-F-030	S11: Server API Resource Management	WNT-01, WNT-11	Aggregate mock/API evidence only: 27 manual checks and Schemathesis coverage 42/42; not mapped per requirement (see API-GATE log)	Test mock: mock-api/server.js	spec/openapi.yaml; logs/api-gate.txt
REQ-F-031	AC-F-031	S11: Server API Resource Management	WNT-01, WNT-11	Aggregate mock/API evidence only: 27 manual checks and Schemathesis coverage 42/42; not mapped per requirement (see API-GATE log)	Test mock: mock-api/server.js	spec/openapi.yaml; logs/api-gate.txt
REQ-F-032	AC-F-032	S11: Server API Resource Management	WNT-01, WNT-11	Aggregate mock/API evidence only: 27 manual checks and Schemathesis coverage 42/42; not mapped per requirement (see API-GATE log)	Test mock: mock-api/server.js	OpenAPI; Addendum 01; API-GATE log
REQ-F-033	AC-F-033	S11: Server API Resource Management	WNT-01, WNT-11	Aggregate mock/API evidence only: 27 manual checks and Schemathesis coverage 42/42; not mapped per requirement (see API-GATE log)	Test mock: mock-api/server.js	spec/openapi.yaml; logs/api-gate.txt

Requirement ID	Acceptance Criterion	BDD Scenario	Ticket	Test	Code	Documentation
REQ-F-034	AC-F-034	S3: Chapter and Book Associations; S4: Symmetric Entry Links; S11: Server API Resource Management	WNT-01, WNT-11	Aggregate mock/API evidence only: 27 manual checks and Schemathesis coverage 42/42; not mapped per requirement (see API-GATE log)	Test mock: mock-api/server.js	spec/openapi.yaml; logs/api-gate.txt
REQ-F-035	AC-F-035	S7: Offline Use and Persistence; S12: Offline Change Synchronization	WNT-01, WNT-12	Aggregate API evidence only: Schemathesis stateful 73 passed; does not verify client queue/retry behavior or each AC assertion	Test mock: mock-api/server.js	spec/openapi.yaml; API-GATE log
REQ-F-036	AC-F-036	S12: Offline Change Synchronization	WNT-01, WNT-12	Aggregate API evidence only: Schemathesis stateful 73 passed; does not verify client persistence/application of changes or each AC assertion	Test mock: mock-api/server.js	spec/openapi.yaml; API-GATE log
REQ-F-037	AC-F-037	S13: Resolve a Synchronization Conflict	WNT-01, WNT-12	Aggregate API evidence only: Schemathesis stateful 73 passed; does not verify client display/selection behavior or each AC assertion	Test mock: mock-api/server.js	spec/openapi.yaml; API-GATE log
REQ-F-038	Not defined in Addendum 02	No dedicated BDD scenario; Addendum 02 requirement	WNT-01, WNT-11	Manual API-GATE checks included registration, login, refresh, and profile; no AC-level test mapping	Test mock: mock-api/server.js	OpenAPI; Addendum 02; API-GATE log
REQ-F-039	Not defined in Addendum 02	No dedicated BDD scenario; Addendum 02 requirement	WNT-01, WNT-11, WNT-12	Manual invalid-token/cross-user checks; sync-isolation evidence is aggregate only; no AC-level test mapping	Test mock: mock-api/server.js	OpenAPI; Addendum 02; API-GATE log

Requirement ID	Acceptance Criterion	BDD Scenario	Ticket	Test	Code	Documentation
REQ-NF-001	AC-NF-001	NFR verification: save latency with 1,000-entry SQLite fixture on minimum OS simulator/emulator	WNT-06			
REQ-NF-002	AC-NF-002	NFR verification: search/filter latency with 1,000-entry SQLite fixture on minimum OS simulator/emulator	WNT-09			
REQ-NF-003	AC-NF-003	S7: Offline Use and Persistence	WNT-06, WNT-09, WNT-10, WNT-12			
REQ-NF-004	AC-NF-004	S7: Offline Use and Persistence	WNT-06, WNT-12			
REQ-NF-005	AC-NF-005	S3: Chapter and Book Associations; NFR verification: relationship integrity across create/edit/detach/delete	WNT-07, WNT-08, WNT-11, WNT-12			
REQ-NF-006	AC-NF-006	S3: Chapter and Book Associations	WNT-08			
REQ-NF-007	AC-NF-007	NFR verification: install and launch on Android API 29+ and iOS 15+	WNT-06			
REQ-NF-008	AC-NF-008	NFR verification: core workflows at 360 x 640 dp viewport	WNT-06			
REQ-NF-009	AC-NF-009	NFR verification: accessibilityLabel presence and platform text scaling	WNT-06			
REQ-NF-010	AC-NF-010	S9: Export Success and Failure	WNT-02, WNT-10			
REQ-NF-011	AC-NF-011	S9: Export Success and Failure	WNT-02, WNT-10			
REQ-NF-012	AC-NF-012	NFR verification: ISO-8601 UTC timestamps and 2 MB FIFO log rotation	WNT-06, WNT-07, WNT-08, WNT-10			
REQ-NF-013	AC-NF-013	NFR verification: confirm no user-facing diagnostic-log UI	WNT-06			
REQ-NF-014	AC-NF-014	NFR verification: TDD evidence and Jest statement/branch coverage thresholds for client and backend components	WNT-05			

Requirement ID	Acceptance Criterion	BDD Scenario	Ticket	Test	Code	Documentation
REQ-NF-015	AC-NF-015	NFR verification: inspect HTTPS transport, JSON payloads, versioned REST paths, HTTP methods, and response status codes	`WNT-01`, `WNT-11`, `WNT-12`	Redocly validates the OpenAPI contract only; HTTPS/wire behavior is not verified by the HTTP localhost mock	Test mock uses HTTP only; not HTTPS evidence	`spec/openapi.yaml`; `logs/api-gate.txt`